Bioresource Technology 454 (2026) 134704

Contents lists available at ScienceDirect

Bioresource Technology

journal homepage: www.elsevier.com/locate/biortech

Synchronously enhanced methanogenesis performance and reduced antibiotic resistance genes of cattle manure anaerobic digestion by air nanobubble water

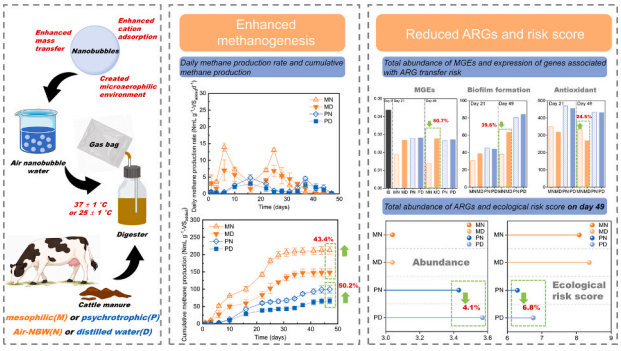
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HIGHLIGHTS

- Air-NBW increased methane yield by 43.4% (mesophilic) and 50.2% (psychrotrophic).
- A 3.92% reduction in ARGs abundance with Air-NBW under psychrotrophic conditions.
- Under mesophilic conditions, Air-NBW reduced the total abundance of MGEs by 50.7%.
- Biofilm formation was inhibited and antioxidant function was enhanced by Air-NBW.
- Air-NBW reduced the ecological risks scores of final digestate.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:
Hydrolysis
Microbial community
Horizontal gene transfer
Mobile genetic elements
Ecological risks

ABSTRACT

The excessive use of antibiotics in livestock farming has made cattle manure (CM) a reservoir for antibiotic resistance genes (ARGs). Anaerobic digestion (AD) can mitigate the risk of spreading ARGs, but its treatment capacity has limitations. This study investigated the effect of air nanobubble water (Air-NBW) on methanogenesis performance and ARGs dynamics during mesophilic and psychrotrophic AD of CM. The results show that Air-NBW enhanced cumulative methane production by 43.4% (mesophilic) and 50.2% (psychrotrophic) by upregulating key functional genes involved in hydrogenotrophic and acetoclastic methanogenesis. Gompertz fitting further showed that the maximum methane yield rate increased by over 62.2% under both temperature conditions. Under mesophilic conditions, Air-NBW had a greater impact on the abundance of mobile genetic elements and the expression of ARG transfer genes than under psychrotrophic conditions. Specifically, Air-NBW led to a 50.7% decrease in mobile genetic elements, a 39.6% downregulation in biofilm-related genes, and a 24.5% upregulation in antioxidant genes, along with a reduction in ARG abundance. This suggests Air-NBW helps lower overall ARG levels. Meanwhile, Air-NBW reduced the abundance of specific ARGs (*lnu(D)*, *lsa(B)*,

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https://doi.org/10.1016/j.biortech.2026.134704
Received 10 February 2026; Received in revised form 21 April 2026; Accepted 21 April 2026
Available online 22 April 2026
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mph(B), *tet(W)*, *tet(T)*, and *tet(Q)*) under both temperature conditions. Ultimately, Air-NBW decreased the ecological risk scores by 3.5% (mesophilic) and by 6.8% (psychrotrophic). In summary, Air-NBW can simultaneously boost methane production and mitigate the ecological risks posed by ARGs, with particularly notable performance under psychrotrophic conditions. Therefore, the addition of Air-NBW to the AD of CM shows great application potential in high-latitude cold regions, serving as an efficient and safe treatment approach.

1. Introduction

China's livestock industry ranks among the world's largest, generating approximately 1.38 billion tonnes of cattle manure (CM) annually (Zhang et al., 2025). The prolonged excessive use of antibiotics during livestock farming has resulted in the appearance of antibiotic resistance genes (ARGs), and the agricultural application of manure further accelerates the environmental spread of ARGs (Liu et al., 2025). ARGs can spread across species through horizontal gene transfer (HGT) facilitated by mobile genetic elements (MGEs), thereby escalating ecological and health risks (Haffiez et al., 2022). By 2050, antibiotic resistance annual deaths worldwide are projected to exceed 10 million (Peng et al., 2025).

Anaerobic digestion (AD) can stably process CM, produce methane, and efficiently convert CM into agricultural fertilizer (He et al., 2025). However, AD may have limited impact on reducing ARGs (Haffiez et al., 2022). During the mesophilic AD of CM, the absolute abundance of 12 ARGs decreased, including *ermF* (Macrolide-lincosamide-streptogramin (MLS) type ARGs), *tetW* (tetracycline type ARGs) and *sul2* (Ma et al., 2019). However, *tetM* and *tetQ* exhibited enrichment under mesophilic and psychrotrophic conditions, while *sul1* exhibited enrichment only under psychrotrophic conditions during the AD of CM (Sun et al., 2016).

Air nanobubble water (Air-NBW) offers the advantages of simple preparation, low cost, and environmental friendliness (Zhang et al., 2024). Air-NBW contains abundant nanobubbles with diameters < 1 μm, which, owing to their high specific surface area, long residence time, and relatively high interfacial charge, can effectively enhance gas dissolution and mass transfer (Bian et al., 2026). This characteristic can promote the hydrolysis of macromolecular organic compounds and further enhance hydrolysis efficiency by creating micro-aerobic environments, optimizing electron transfer, and enriching facultative anaerobic bacteria (Chuenchart et al., 2021). As a result regulates the production and composition of volatile fatty acid (VFAs), mitigates the risk of acid accumulation, enhances system stability, and ultimately increases methane production (Hou et al., 2025). Additionally, Air-NBW can scavenge reactive oxygen species (ROS) by enhancing the activity and gene expression of antioxidant enzymes such as glutathione peroxidase, superoxide dismutase, and catalase (Zhang et al., 2022).

In sludge AD, Air-NBW increased methane production in the anaerobic digestion process, with the 40.0% NBW group showing a 16.0% increase in methane yield, likely due to the promotion of the hydrolysis stage, increased VFA accumulation, and the enhancement of the microbial growth environment through nanobubbles (Wang et al., 2025). During corn stover AD, Air-NBW enhanced methane production by 11.9% to 24.1% through promoting microbial growth for lignocellulose degradation, improving hydrolysis efficiency, and stabilizing the anaerobic environment via enhanced mass transfer (Bian et al., 2026). During the mesophilic AD of CM, Air-NBW increased the abundance of key hydrolytic bacteria (*Bacillus*) and mixotrophic methanogens (*Methanosarcina*) from 0.3% and 20.3% to 39.2% and 47.4%, respectively, and cumulative methane production by 9.7% (Zhang et al., 2024). While substantial evidence supports the positive effect of Air-NBW on methane production in various AD systems, little to no research has been conducted on its impact on ARGs.

The reductive effect of NBW on ARGs results from the combined effects of substrate environment changes, genetic material transfer, and community restructuring. Air-NBW may promote hydrolysis and methane production under mildly aerobic conditions or redox regulation, potentially reducing VFA accumulation, restructuring the anaerobic system, and weakening the competitive advantage of high ARG bacteria, which could, in turn, alleviate ROS-related stress responses and HGT (Chuenchart et al., 2021; Feng et al., 2025). Biofilm facilitates the HGT of ARGs through MGEs such as plasmids and integrons, making them recognized hotspots for ARG dissemination (Michaelis & Grohmann, 2023). NBW may disrupt the biofilm matrix, which is primarily composed of extracellular polymeric substances (EPS), through nano-scale gas–liquid interface effects, thereby affecting its stability (Kobayashi et al., 2025). Given the critical role of EPS in cell adhesion, aggregation, and biofilm stability, the disruption of EPS and biofilm by NBW could reduce cell aggregation, leading to changes in community structure (Wang et al., 2024). These microbial composition changes may impact the abundance and host range of ARGs and MGEs, ultimately reshaping the antibiotic resistance gene profile (Michaelis & Grohmann, 2023). Specifically, the effects of Air-NBW on the abundance of MGEs, the expression of functional genes related to biofilm formation, antioxidative defense, DNA repair and recombination, and the type IV secretion system remain unclear. Furthermore, the potential links between these factors and the HGT of ARGs are not well understood. Additionally, the role of Air-NBW in attenuating ARGs and mitigating ecological risks has been largely overlooked.

This study investigates the effects of adding Air-NBW on system performance during mesophilic and psychrotrophic AD of CM. Further analysis examined the regulatory role of Air-NBW in ARGs transfer, degradation, and ecological risk.

2. Materials and methods

2.1. Materials

CM was collected from a standardized dairy farm in Lanzhou, China, and stored at 4 °C. The median particle size (D50) of the CM was 33.1 μm (Fig. S1). Seed sludge for each digester was waste activated sludge predigested anaerobically for 10 days under either mesophilic (37 ± 1 °C) or psychrotrophic (25 ± 1 °C) conditions. Air-NBW was prepared by injecting air into distilled water (0.4–0.5 MPa, 20 min, 20 °C) using a nanobubble generator (JXWNP-AL01, Jinxiang, China) with continuous recirculation. Fresh NBW appeared milky white and became clear within 5–10 min as larger bubbles rose and dissipated. The physicochemical characteristics of CM and digested sludge (Table S1), as well as the basic properties of Air-NBW, are provided in Table S2 and Fig. S2.

2.2. Experimental setup

In each digester, CM was mixed with the water (Air-NBW or distilled water) at a 1:1 mass ratio with a total mass of 1200 g (Zhang et al., 2024). Each digester was then inoculated with 150 g of anaerobic digested sludge, establishing the initial substrate (IS). The inoculum-to-substrate ratio (ISR) and initial substrate concentration on a volatile solids (VS) basis (C_0, VS) were 4.8×10^{-2} and 55.9 ± 0.7 g VS/L, respectively (mesophilic), and 5.3×10^{-2} and 56.2 ± 0.7 g VS/L, respectively (psychrotrophic). Batch experiments were run under mesophilic or psychrotrophic conditions, using NBW as the treatment and distilled water as the control, yielding four groups: the mesophilic nanobubble water group (MN), the mesophilic distilled water group (MD), the psychrotrophic nanobubble water group (PN), and the psychrotrophic distilled water group (PD). Each of the four experimental conditions (MN, MD, PN, and PD) was set up in three parallel replicates.

A 2 L bottle reactor was sealed with a custom lid fitted with two 4 mm metal tubes connected to silicone hoses. One tube was connected to a gas collection bag for biogas storage, and the other was used to withdraw digestate via a peristaltic pump. The reactor was shaken for 1 min daily for mixing. Digestate was sampled on days 0 and 3, then every 7 days until day 49.

2.3. Anaerobic digestion analysis methods

Uniformly mixed samples were collected regularly from each digester, centrifuged, and filtered through PES membrane with a 0.45 μm aperture, yielding the filtrate. Total solids (TS) and volatile solids (VS) were analyzed from the original samples of cattle manure and digested sludge, while pH, soluble chemical oxygen demand (SCOD), and total ammonia nitrogen (TAN) were measured from the filtrate, all according to standard methods (APHA, 1999). Free ammonia nitrogen (FAN) was calculated using Hansen's formula (Hansen et al., 1998). VFAs and methane content of biogas were determined using a gas chromatography (7890Plus, Ruihong, China). The biogas volume was determined by a gas flow meter (LMI-2, Alpha, China). Elemental composition (C, H, N, S) was determined using an organic elemental analyzer (Vario EL Cube, Elementar, Germany).

2.4. Microbiome and macrogenetic analysis

Sample DNA was extracted using the HiPure Soil DNA kit at predetermined time points and analyzed the 16S rDNA gene through high-throughput sequencing. Bacterial and archaea amplification was performed using primer sets (341F: CCTACGGGNGGCWGCAG and 806R: GGACTACHVGGGGTATCTAAT) and (Arch519F: CAGCMGCCGCGGTAA and Arch915R: GTGCTCCCCCGCCAATTCCT), respectively. The resulting amplicons were subjected to paired-end sequencing on the Illumina PE250 platform, followed by comprehensive bioinformatic processing.

Metagenomic sequencing using the Illumina NovaSeq platform, combined with bioinformatics analysis, was used to compare functional gene profiles of samples on day 0, day 21, and day 49 (Methods S1) (Pärnänen et al., 2018; Yin et al., 2023). Ecological risk scores (ERS) were calculated using the MetaCompare 2.0 framework to assess antibiotic resistance risk. Metagenomic reads were assembled into contigs, annotated for ARGs, MGEs, and host taxonomy, and the co-localization of ARGs with MGEs and pathogenic hosts was identified. Abundances were normalized based on contig counts. Finally, samples were projected into a hazard space to derive and rank resistome risk scores (Rumi et al., 2024).

2.5. Data analysis

Statistical analysis was performed using SPSS 27.0 (Windows version). Differences between experimental groups were compared via one-way analysis of variance (ANOVA) (significance threshold $p < 0.05$). All data are expressed as mean ± standard deviation. The cumulative methane production data were fitted using the modified Gompertz equation, and the model parameters were estimated by nonlinear least-squares regression (Methods S2) (Dhamodharan et al., 2015).

3. Results and discussion

3.1. Anaerobic digestion performance and methane yield

3.1.1. pH, SCOD, TAN, FAN and VFAs

AD relies on maintaining pH within an optimal range to support microbial growth and metabolic activity, thereby ensuring process stability and efficiency (Chen et al., 2022). The pH dropped to its lowest point on day 3 (mesophilic) and day 7 (psychrotrophic) (Fig. 1a). It

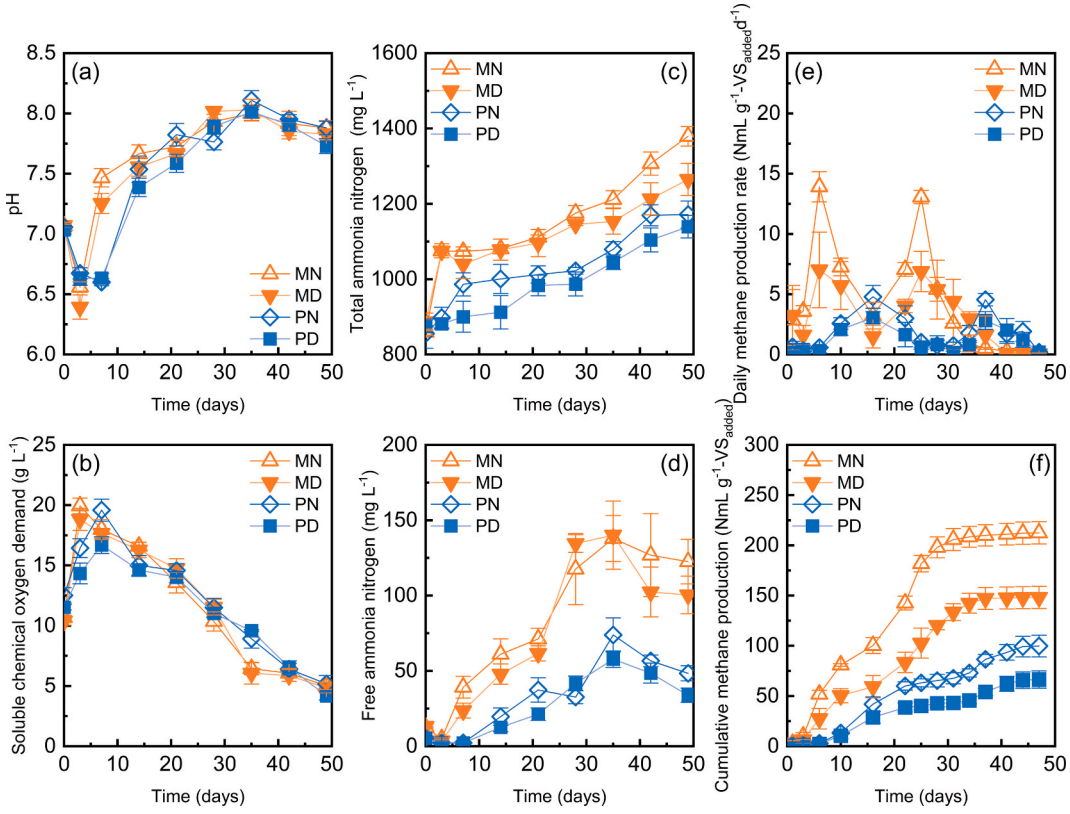


Fig. 1. Changes in pH (a), soluble chemical oxygen demand (b), total ammonia nitrogen (c), free ammonia nitrogen (d), daily methane production rate (e), and cumulative methane production (f) during anaerobic digestion.

subsequently rebounded, reaching a peak before experiencing a second decline on day 35. The initial pH drop was associated with the accumulation of VFAs, which peaked on day 3 and day 7, respectively (Fig. S3) (Bian et al., 2026). Subsequently, ammoniacal nitrogen contributed partial alkalinity and buffering capacity, mitigating VFA accumulation and maintaining the reactor pH within a mildly alkaline to neutral range (Li et al., 2024).

SCOD peaked on day 3 (mesophilic) and day 7 (psychrotrophic), with higher values observed in the NBW group compared with the control, followed by a gradual decline (Fig. 1b). SCOD variations reflect hydrolysis-driven VFA production (Bian et al., 2026). The syntrophic oxidation of propionate and butyrate is thermodynamically constrained, requiring extremely low hydrogen or formate partial pressures (Lv et al., 2020). Air-NBW-induced microaerophilic conditions stimulate facultative anaerobes to scavenge oxygen and maintain redox balance, thereby alleviating thermodynamic limitations (Chuenchart et al., 2021). This enhances syntrophic cooperation, electron transfer, and accelerates VFA conversion, contributing to higher SCOD peaks. The subsequent decline in SCOD is attributed to the consumption of organic acids by methanogenic archaea, which convert them into methane (Zhang et al., 2024).

TAN is a product of the degradation of nitrogenous organic matter in AD (Gao et al., 2023). TAN exhibited an increasing trend across all groups, with higher levels in the NBW group compared to the corresponding control under both temperature conditions (Fig. 1c). Air-NBW enhanced substrate hydrolysis and microbial activity, accelerating the degradation of organic nitrogen in cow manure and increasing TAN levels, with total concentrations remaining below 1500 mg L^{-1}, which is within a suitable range for microbial use (Peng et al., 2025; Zhang et al., 2024). FAN showed an initial increase followed by a decrease, peaking on day 35 (Fig. 1d). The effect of Air-NBW on FAN mirrored its influence on pH. Under alkaline conditions, ammonia primarily exists as free ammonia, and changes in water pH directly affect the balance between ammonia and ammonium ions. FAN remained below 200 mg L^{-1} in all treatments and did not significantly affect methane production (Zhang et al., 2024).

3.1.2. Methane yield

Daily methane production rate is a key indicator for evaluating anaerobic digestion efficiency (Chen et al., 2022). It exhibited two peaks under both temperature conditions, with higher values observed in the NBW group compared with the corresponding controls (Fig. 1e). The first peak occurred on day 3 (mesophilic) and day 7 (psychrotrophic), while the second peak was observed on days 21 and 35, respectively, followed by a gradual decline after each peak. During the initial phase, Air-NBW may boost AD start-up by transiently creating a microaerophilic niche that speeds oxygen removal and strict anaerobiosis, improves mass transfer and hydrolysis, and may promote syntrophy mediated by direct interspecies electron transfer (DIET), thereby reducing VFA accumulation and increasing methane yield (Bian et al., 2026; Chuenchart et al., 2021; Lv et al., 2020; Zhang et al., 2024). At the VFA peak, acetate predominated, indicating substrate accumulation toward subsequent methanogenesis (Fig. S3) (Bian et al., 2026). Subsequently, propionate exceeded acetate and became the primary inhibitor of methane production due to limited conversion, leading to a gradual decline in daily methane yield (Fu et al., 2020). On day 21 under mesophilic conditions, *Bacillota* and *Bacteroidota* predominated, with a larger increase in *Bacteroidota* in MN than in MD (Fig. S4a). *Bacteroidota* shows strong cellulose-degrading capacity during AD, likely due to its electroactivity and higher glycolytic enzyme activity than *Bacillota* (Sha et al., 2023). On day 35 under psychrotrophic conditions, the relative abundance of *Methanosarcina* in PN was higher than in PD (Fig. S4b). *Methanosarcina* can accelerate VFAs utilization and methane synthesis through interspecies DIET (Liu et al., 2023). Air-NBW may enhance the recovery phase by reducing H_2 and formate diffusion limitations and promoting DIET, thereby improving VFAs utilization and methane performance during recovery (Akram et al., 2024; Wang et al., 2020). Accordingly, cumulative methane production was higher in MN than in MD, and in PN than in PD, by 43.4% and 50.2%, respectively (Fig. 1f).

The Gompertz model provided a good fit to the cumulative methane production profiles (Table 1). Rm (maximum methane yield rate) and UMP (ultimate methane production) can both be used to characterize methane production performance in AD, with Rm reflecting the methane production rate and kinetic characteristics of the system and UMP representing its ultimate methane production potential (Lee et al., 2024). In the NBW group, both Rm and UMP were significantly higher than those in the corresponding control groups ($p < 0.05$). These results suggest that Air-NBW may enrich hydrolytic-acidogenic bacteria and syntrophs, enhancing H_2 and formate transfer within flocs to ease thermodynamic constraints and boost methanogenic synergy (Akram et al., 2024; Hou et al., 2025; Wang et al., 2024). The magnitude of the λ directly reflects the speed at which microorganisms mount an effective response to environmental stress. A higher λ prolongs the activation time of microbial enzyme systems and the degradation cycle of environmental inhibitory substances, thereby delaying the exponential growth phase and postponing the initiation of methane production (Fu et al., 2020). Air-NBW shortened λ under mesophilic conditions, yet lengthened λ under psychrotrophic conditions. Under mesophilic conditions, Air-NBW may significantly enhance the degradation of macromolecular organic compounds by strengthening the electron transport system and optimizing the catalytic efficiency of enzymes associated with hydrolysis and acid production (Bian et al., 2026). However, under psychrotrophic conditions, anaerobic microorganisms exhibit inherently low activity, and adding Air-NBW further alters the internal environment of the digester, thus requiring them to undergo an adaptation period (Wang et al., 2020). In summary, Air-NBW significantly enhanced methanogenesis performance during mesophilic and psychrotrophic AD of CM.

3.2. Bacterial and archaeal communities and methanogenic pathways

At the phylum level of bacterial communities, *Bacillota*, *Bacteroidota*, *Spirochaetota*, *Chloroflexota*, and *Synergistota* were the key phyla dominating lignocellulose polymer degradation and fermentation processes across all groups (Fig. 2a) (Lu et al., 2022). On day 21, Air-NBW increased the overall relative abundance of these five phyla under mesophilic conditions, from 54.1% in MD to 54.6% in MN. This change suggests accelerated organic matter breakdown via enrichment of hydrolytic bacteria and consequently enhanced conversion to VFAs and methane (Figs. S3 and 1f) (Hou et al., 2025). In contrast, under psychrotrophic conditions, Air-NBW reduced their combined abundance relative to the corresponding control, consistent with the delayed onset of gas production. On day 49, the NBW group showed a higher combined abundance of the five dominant bacterial phyla than the corresponding

Table 1
Parameters of the modified Gompertz model.

	Lag phase, λ (d)	Maximum methane yield rate, R_m (NmL g^{-1} VS_{added} d^{-1})	Ultimate methane production, UMP (NmL g^{-1} VS_{added})	R^2
MN	2.0 ± 0.4^a	8.5 ± 0.4^a	225.0 ± 12.9^a	0.98 ± 0.00
MD	2.6 ± 1.5^a	5.0 ± 0.2^b	167.6 ± 16.4^b	0.98 ± 0.00
PN	5.3 ± 1.0^b	3.2 ± 0.3^c	108.4 ± 15.1^c	0.98 ± 0.01
PD	4.7 ± 0.8^b	1.9 ± 0.2^d	77.2 ± 20.7^d	0.97 ± 0.01

Note: All values are expressed as mean ± standard deviation (n = c. Methane production was normalized to standard temperature and pressure (STP; 273 K and 1 atm). The letters a, b, c, and d were assigned according to Duncan's test. When there were no duplicate labeling symbols between the two groups, they were considered to be significantly different at the $p < 0.05$ level.

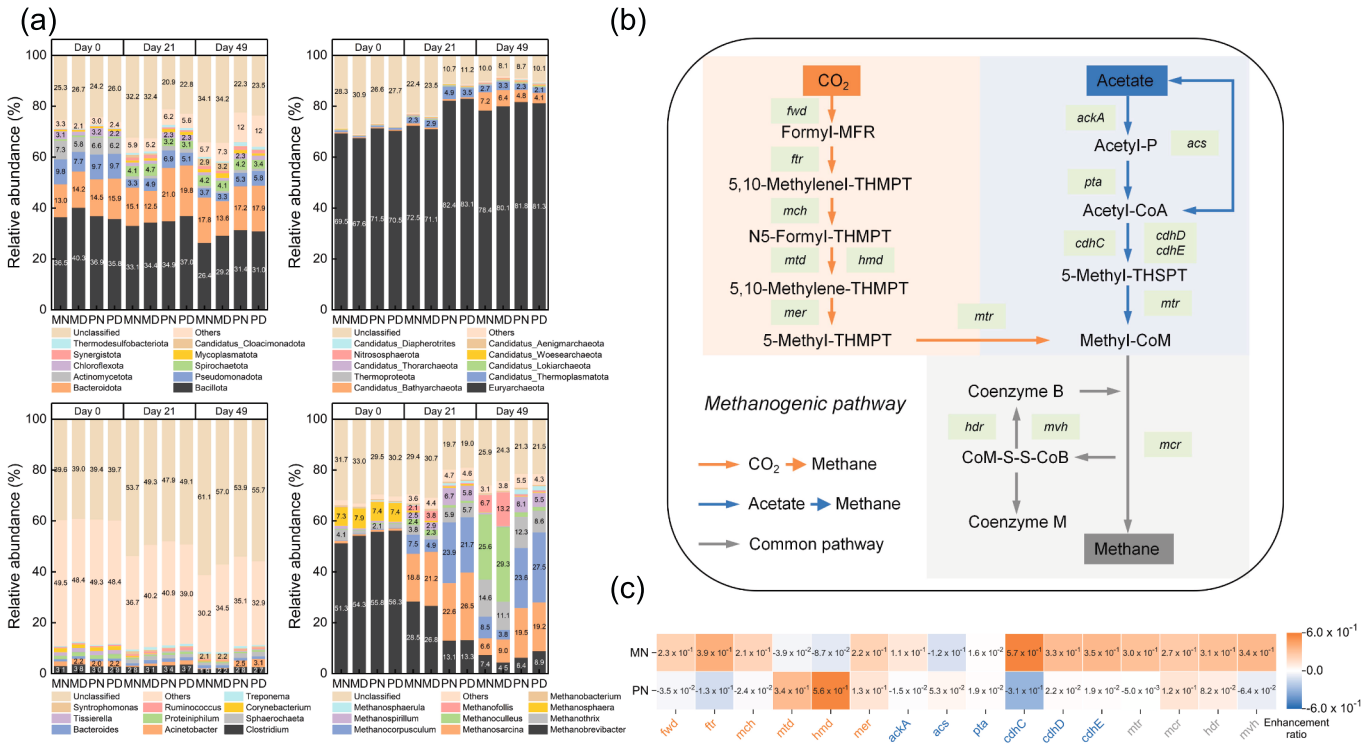


Fig. 2. Changes in the relative abundances of bacterial and archaeal communities in different digesters (a, bacterial phylum level; b, bacterial genus level; c, archaeal phylum level; d, archaeal genus level), and the main methanogenic pathways (hydrogenotrophic, M00567; acetoclastic, M00357) in anaerobic digestion (e), and the heat map showing differences in the expression of methanogenesis-related functional genes between the experimental and control groups on day 21 (f).

control under both temperature conditions, rising from 49.0% in MD to 50.8% in MN (mesophilic) and from 55.2% in PD to 56.5% in PN (psychrotrophic). Genus-level analysis identified *Clostridium* as dominant. Under mesophilic conditions, it rapidly converts complex organics into VFAs, H_2 and CO_2 to support methanogenesis, whereas under psychrotrophic conditions, conversion is slower due to reduced enzymatic and metabolic activity, often requiring acclimation and longer retention time or lower loading for stability (Nie et al., 2021). Further genus-level analysis revealed that the NBW group exhibited higher relative abundances of taxa classified as "*Others*" and "*unclassified*," alongside an elevated Chao1 index (Fig. 2b and S5). The addition of Air-NBW increased microbial richness and community complexity, potentially through enhanced syntrophic interactions and community restructuring (Hou et al., 2025; Zhang et al., 2024). These results suggest that Air-NBW may promote the enrichment of the key phyla dominating lignocellulose polymer degradation and fermentation during AD, thereby enhancing the hydrolysis and acidification of organic matter. This facilitated increased VFAs production and supplied more substrates for methanogenesis, ultimately leading to higher methane yields in the NBW group compared with the corresponding control.

At the phylum level of archaeal communities, *Euryarchaeota* (67.6–80.1%) exhibited overwhelmingly dominated (Fig. 2c). This phylum primarily comprises hydrogenotrophic methanogens (*Methanobrevibacter*, *Methanocorpusculum*, *Methanoculleus*, *Methanospirillum*, *Methanofollis*, *Methanosphaerula* and *Methanobacterium*), acetoclastic methanogens (*Methanothrix*), and mixotrophic methanogens (*Methanosarcina*) (Fig. 2d) (Bian et al., 2026; Peng et al., 2025). Methanoculleus is a hydrogenotrophic methanogen that reduces CO_2 with H_2 to keep H_2 partial pressure low and sustain syntrophic methane formation, while Methanothrix is more competitive at low acetate concentrations (Kakuk et al., 2021; Wang et al., 2026). Temperature strongly regulates methanogenesis: mesophilic conditions boost Methanoculleus activity and biogas yield, whereas psychrotrophic conditions slow start-up and methane production, favoring Methanothrix at low acetate (Rama et al., 2024). At the genus level, on day 21, the persistent presence of Air-NBW may have created a micro-oxic niche that favored hydrogenotrophic methanogens capable of remaining metabolically active under harsh conditions, thus gaining a competitive advantage (Peng et al., 2025). Accordingly, their relative abundance increased from 41.4% in MD to 43.7% in MN and from 43.6% in PD to 46.5% in PN. By day 30, the system had entered a strictly anaerobic state; as oxygen was depleted, the microbial community gradually became dominated by acetoclastic methanogens (Chuenchart et al., 2025). On day 49, their relative abundance increased from 11.1% in MD to 14.6% in MN, and from 8.6% in PD to 12.3% in PN.

The methanogenesis pathways in the AD system is primarily dominated by hydrogenotrophic methanogenesis (M00567) and acetoclastic methanogenesis (M00357) (Fig. 2e) (Gao et al., 2023). The ratio of homoacetogenesis to combined hydrogenotrophic and acetoclastic methanogenesis ranged from 19.3% to 25.3%, consistently below 25.5%, indicating a limited contribution. NBW showed notable differences under mesophilic conditions but minimal variation under psychrotrophic conditions. Under both temperature conditions, adding Air-NBW induced the upregulation of functional genes associated with the hydrogenotrophic methanogenesis pathway (Fig. 2f). Compared with PD, PN showed higher *hmd* expression ($\log_2FC = 5.7 \times 10^{-1}$), while compared with MD, *ftr* expression was higher in MN ($\log_2FC = 3.9 \times 10^{-1}$). This may be due to Air-NBW enriching the key hydrogenotrophic methanogens *Methanocorpusculum* and *Methanobrevibacter* under psychrotrophic and mesophilic conditions, respectively, thereby enhancing their utilization of H_2 and CO_2, promoting a mutualistic relationship with fermentative bacteria, alleviating product inhibition, and thus facilitating methane production in the system (Rama et al., 2024). In the acetoclastic methanogenesis pathway, the *cdhC* ($\log_2(FC) = 5.7 \times 10^{-1}$) and *cdhE* ($\log_2(FC) = 3.5 \times 10^{-1}$) gene in MN showed higher expression levels compared with MD. This is consistent with the earlier observation that acetate production was higher in MN than in MD (Fig. S3a and S3b). The nanobubble interface may enhance mass transfer efficiency for

acetate (Hou et al., 2025). Under psychrotrophic conditions, *cdhC* gene expression was significantly downregulated in PN compared with PD. In VFA-limited environments, microbial communities allocate resources toward the hydrogenotrophic methanogenesis pathway, which offers higher energy returns (Liu et al., 2020). Air-NBW increased the proportion of hydrogenotrophic methanogen on day 21 (Fig. 2d). Therefore, daily methane production rate by PN is greater than that of PD (Fig. 1e).

Overall, under psychrotrophic conditions, Air-NBW primarily promotes the hydrogenotrophic methanogenesis pathway, whereas under mesophilic conditions, both hydrogenotrophic and acetoclastic methanogenesis pathways are enhanced. This temperature-dependent modulation of methanogenic pathways directs metabolic fluxes toward the energetically favorable pathways in each environment, thereby facilitating more efficient methane production.

3.3. Changes in antibiotic resistance genes and mobile genetic elements

MLS and tetracycline ARGs were predominant in the AD system of CM (Fig. 3a and S6). The NBW group showed a reduction in the total number of ARGs by 0.1% (mesophilic) and 4.1% (psychrotrophic) on day 49 compared with the corresponding control group. For MLS and tetracycline types ARGs, Air-NBW suppressed the abundance of *lnu(D)*, *lsa(B)*, *mph(B)*, *tet(W)*, *tet(T)*, and *tet(Q)* under both temperature conditions on days 21 and 49 (Fig. 3b and 3c). The reductions in the six genes were approximately 5.1–15.6%, 0.0–13.7%, 14.2–73.6%, 1.2–14.0%, 13.0–41.5% and 4.9–49.2%, respectively. *Tet(W)*, *tet(T)*, and *tet(Q)* belong to the ribosomal protection proteins gene family mediating tetracycline resistance in Gram-positive bacteria (Zhanel et al., 2016). The relative abundance of Gram-positive bacteria in the NBW group was lower than that in the corresponding control group. *Lnu (D)*, *mph(B)* and *tet(T)* showed significant negative correlations with the concentration of acetic acid and propionic acid ($p < 0.05$), with the following correlation coefficients: *Lnu(D)* (acetic acid -6.5×10^{-1}, propionic acid -6.1×10^{-1}), *mph(B)* (acetic acid -6.8×10^{-1}, propionic acid -6.7×10^{-1}), and *tet(T)* (acetic acid -6.1×10^{-1}, propionic acid -5.9×10^{-1}) (Fig. S7a and S7c). Despite the stage-dependent consumption of VFAs, Air-NBW still showed a clear promoting effect on VFAs production (Fig. S3). In line with this, *lsa(B)* and *tet(W)* exhibited significant negative correlations with TAN and FAN, with correlation coefficients as follows: *lsa(B)* (TAN -7.1×10^{-1} ($p < 0.01$), FAN -7.1×10^{-1} ($p < 0.05$)), and *tet(W)* (TAN -8.4×10^{-1}, FAN -9.2×10^{-1}) ($p < 0.001$) (Fig. S7a and S7c). Additionally, *Lnu(D)* and *tet(Q)* showed significant negative correlations with the bacterial phyla *Pseudomonadota* and *Actinomycetota*, with correlation coefficients as follows: *Lnu(D)* (*Pseudomonadota* -8.0×10^{-1}, *Actinomycetota* -7.7×10^{-1}) ($p < 0.01$), and *tet(Q)* (*Pseudomonadota* -8.3×10^{-1} ($p < 0.001$), *Actinomycetota* -5.9×10^{-1} ($p < 0.05$)) (Fig. S7b and S7d). Air-NBW increased the total abundance of *Pseudomonadota* and *Actinomycetota*, two phyla possessing hydrolytic capabilities (Peng et al., 2025). In summary, *lnu(D)*, *lsa(B)*, *mph(B)*, *tet(W)*, *tet(T)*, and *tet(Q)* all showed negative correlations with hydrolytic activity in AD, consistent with the phenomenon that Air-NBW enhances hydrolysis. On days 21 and 49, Air-NBW increased the abundance of *lnu(H)*, *lnu(B)*, *vat(B)*, *tet(M)* and *tet(44)* (Fig. 3b and 3c). Their risk levels were IV, I, IV, II, and III, respectively (Peng et al., 2025; Zhang et al., 2021). The risks associated with *lnu(H)*, *vat(B)* and *tet(44)* were relatively low. Although Air-NBW led to a slight increase in *lnu(B)*, the change was minimal. In contrast, *tet(M)* exhibited an overall decreasing trend. In conclusion, while some genes showed an increase in risk, their overall impact remained negligible.

On days 21 and 49, the total abundance of MGEs in the NBW group was lower than in the control (Fig. 4a). Under mesophilic conditions, MN decreased by 30.2% compared with MD, and by 50.7%, respectively. In contrast, under psychrotrophic conditions, PN decreased only slightly relative to PD, by 1.4% and 2.2%. This decrease coincided with Air-NBW-induced shifts in key environmental parameters, including higher pH and TAN (Fig. 1a and 1c), and with concurrent changes in community composition. In particular, the relative abundances of *Bacillota* and *Pseudomonadota*, which represent major potential MGE-associated hosts in this system, were reduced in the NBW group (Fig. 2a) (Flores-Orozco et al., 2024; Liu et al., 2025; Zhang et al., 2020). However, under the psychrotrophic conditions, the strong constraints imposed by temperature on microbial metabolism and growth slowed community succession, thereby markedly weakening the impact of environmental changes, such as pH and TAN, on host populations and their associated mobile genetic elements (Dal Bello & Abreu, 2024). The relationship between MGE reductions and host bacterial shifts is crucial (Figs. S8 and S9). The contributions of individual metagenome-assembled genomes bins to the treatment-associated differences in total MGE abundance (Air-NBW vs. control) were assessed on days 21 and 49 (Fig. 4b). The cumulative contribution of the top 10 metagenome-assembled genomes bins decreased from 1.9×10^{-2} on day 21 to 7.0×10^{-3} on day 49, compared with a decline from 5.0×10^{-3} to 4.2×10^{-3} under psychrotrophic conditions. These findings demonstrate that the ΔContribution was higher under mesophilic conditions, particularly on day 21, and decreased over time under both temperature conditions. Therefore, Air-NBW had a more pronounced effect on reducing MGEs under mesophilic conditions.

The HGT of ARGs is associated with several key pathways, including biofilm formation, antioxidation, DNA repair and recombination, as well as the type IV secretion system (T4SS) (Fig. S10) (Peng et al., 2025; Tang et al., 2023). Biofilms serve as a critical barrier for microorganisms

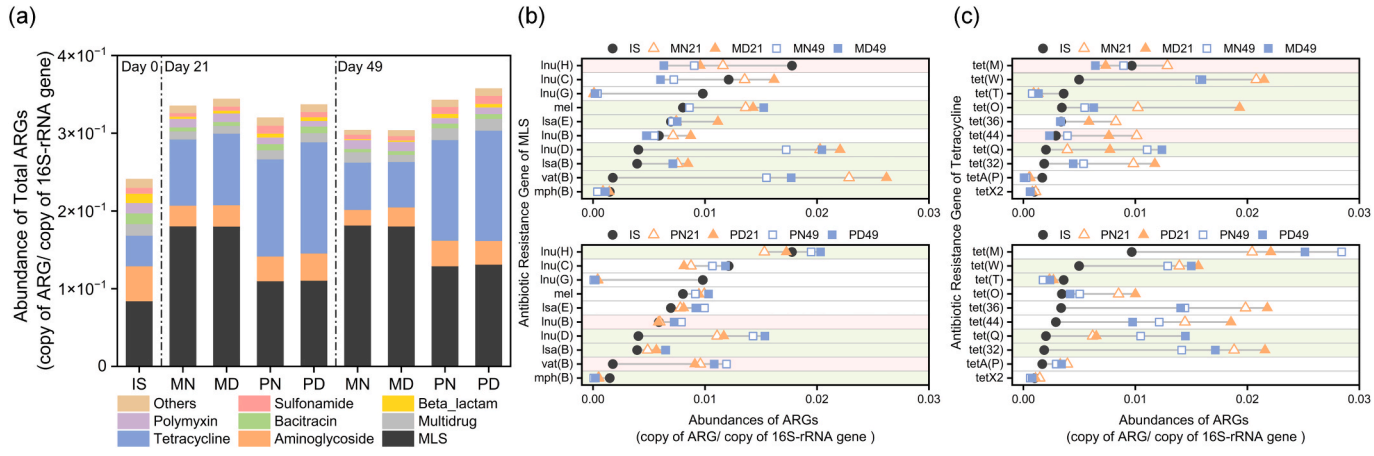


Fig. 3. Changes in the total abundance of antibiotic resistance genes (a), and macrolide-lincosamide-streptogramin antibiotic resistance genes (b), and tetracycline antibiotic resistance genes (c) in anaerobic digestion.

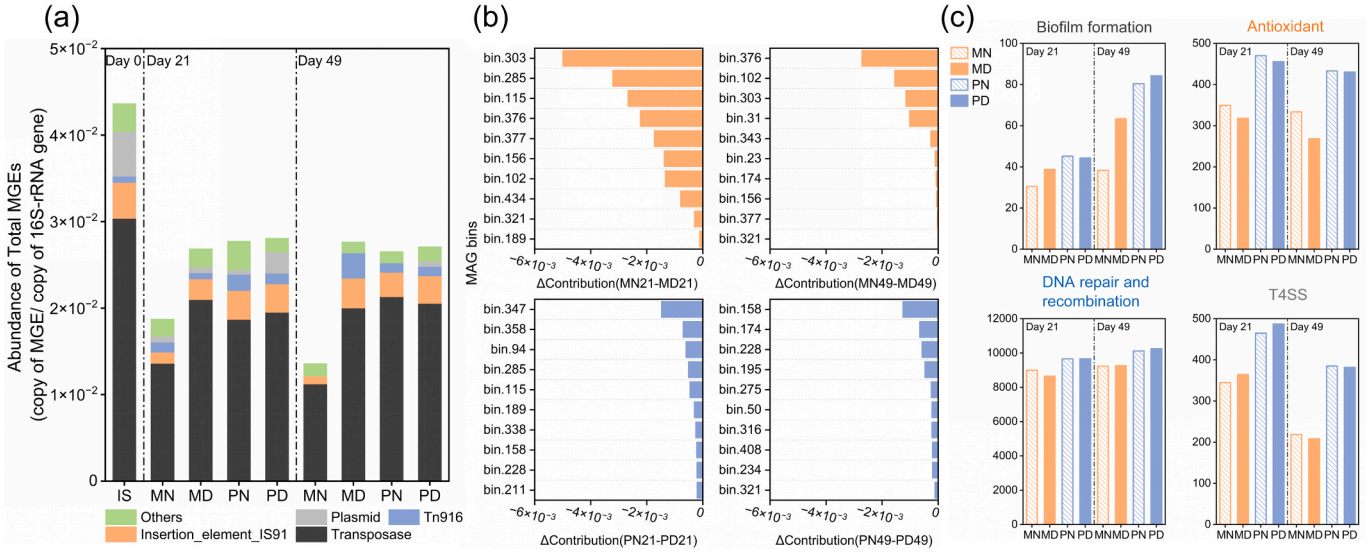


Fig. 4. Changes in the total abundance of mobile genetic elements (a), and top 10 metagenome-assembled genomes bins driving the difference in ΔContribution (host-level drivers of air nanobubble water effects) (b), and expression of functional genes related to biofilm formation, antioxidant, DNA repair and recombination, and the type IV secretion system in different digesters (c).

against external stress, with their formation dependent on the secretion of EPS (Fig. 4c) (Tang et al., 2023). Increased EPS production enhances cell adhesion and may promote the transfer of ARG genes under high redox conditions (Wang et al., 2024). Air-NBW reduces ARG transmission by downregulating EPS synthesis genes, disrupting EPS barriers primarily composed of polysaccharides, destabilizing the biofilm, reducing cell aggregation, and decreasing donor-receptor proximity (Kobayashi et al., 2025; Wang et al., 2024). Specifically, on day 49, MN exhibited a 39.6% decrease compared with MD, whereas PN showed only a 4.6% decrease relative to PD. Under antibiotic stress, microorganisms may activate antioxidant defense systems to scavenge excess ROS, thereby alleviating oxidative DNA damage and limiting ROS-mediated HGT of ARGs (Peng et al., 2025). Air-NBW may enhance antioxidant activity by expanding the gas–liquid interfacial area, boosting dissolved O_2 transfer, and stabilizing interfacial charge, thereby activating redox-responsive regulation and accelerating ROS scavenging (Wang et al., 2020; Zhang et al., 2022). On day 49, antioxidant-related functional genes were upregulated, with MN showing a 24.5% increase compared with MD, whereas PN exhibited a relatively minor increase of 0.6% relative to PD (Fig. 4c). DNA damage can trigger the SOS response, leading to increased expression of recombination and repair genes and potentially enhancing HGT of ARGs (Yang et al., 2022). Air-NBW treatment was accompanied by a reduction in the expression of genes involved in DNA repair and recombination (Fig. 4c). On day 49, MN and PN decreased by 0.3% and 1.2% compared with MD and PD, respectively. T4SS efficiently mediates conjugative transfer through coordinated action of multiple protein components (Tang et al., 2023). The expression levels of T4SS-related functional genes in the NBW group were similar to those in the corresponding control group (Fig. 4c). Overall, the observed changes in these

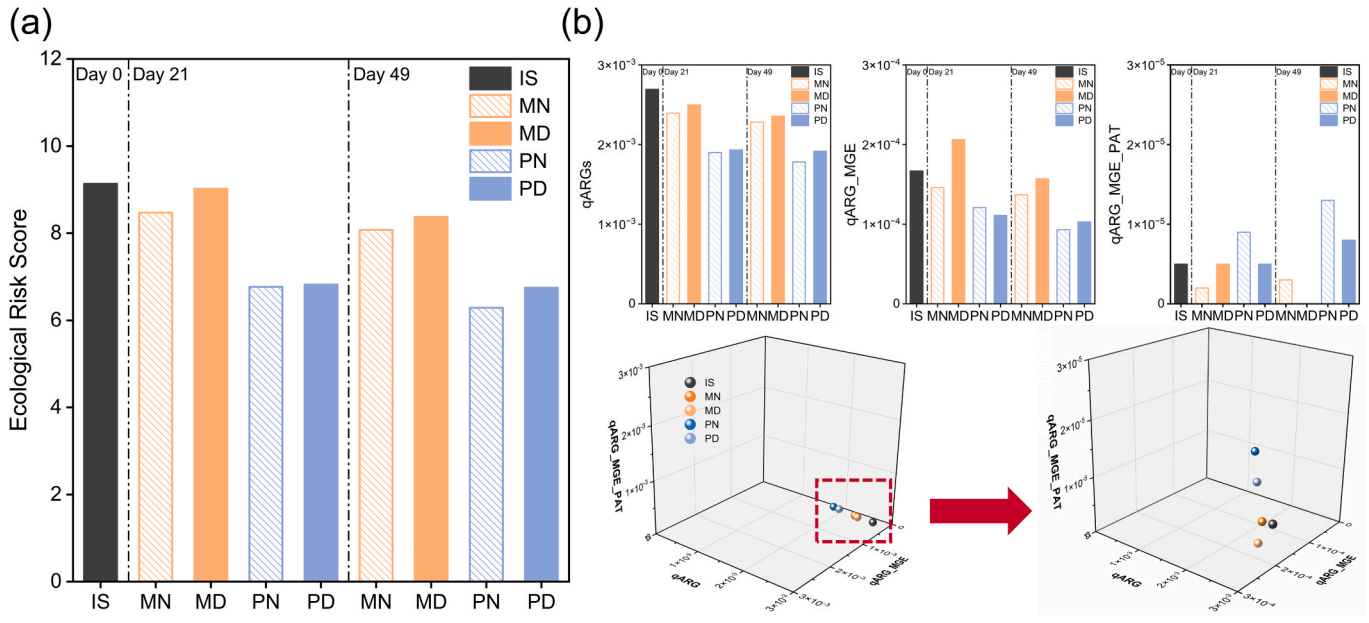


Fig. 5. The ecological risk scores for antibiotic resistance genes in different digesters based on MetaCompare 2.0 (a), and qARG, qARG_MGE, qARG_MGE_PAT, and a three-dimensional scatter plot constructed from these three indicators (b).

functional pathways suggest that Air-NBW reduces the potential for ARGs dissemination via HGT.

3.4. Ecological risks of antibiotic resistance genes

Compared with the IS, ERS were reduced across all treatment groups, with further mitigation observed after adding Air-NBW (Fig. 5a). The conjugation of ARGs with MGEs may be the core mechanism driving the spread of ARGs in AD (Jiao et al., 2024). The co-occurrence level of ARGs and MGEs in the NBW group was lower than that in the corresponding control group, indicating reduced mobility (Fig. 5b). On day 49, MN and PN decreased by 12.7% and 9.7% compared with MD and PD, respectively. The NBW group was closer to the origin compared with the corresponding control group, indicating that Air-NBW can reduce ERS (Rumi et al., 2024). Additionally, the ERS under psychrotrophic conditions was lower than that under mesophilic conditions. On day 49, PN and PD decreased by 22.2% and 19.4% compared with MN and MD, respectively. This may be because, under mesophilic conditions, the proportion of drug-resistant bacteria and genetic variability within populations increase relative to psychrotrophic conditions, raising the probability of drug-resistant mutations occurring (Rodríguez-Verdugo et al., 2020). Furthermore, as reservoirs of ARGs, microbial communities amplify the effect of temperature on the spread of resistance through internal gene transfer (Luo et al., 2025).

In practical applications, AD has been widely adopted for treating CM, efficiently degrading organic waste into biogas for energy recovery while reducing pollutant emissions (He et al., 2025). As a green additive, Air-NBW effectively enhances the system's hydrolysis efficiency and methane production without introducing additional contaminants (Bian et al., 2026). It also improves digestate quality, enabling treated CM to serve as a soil conditioner. This method avoids secondary environmental hazards while offering high economic viability, facilitating resource recovery, and advancing green, low-carbon agricultural transformation. Furthermore, Air-NBW demonstrates potential advantages in controlling ARGs, helping reduce overall ERS, particularly under psychrotrophic conditions where risk mitigation is more pronounced. Consequently, it is well-suited for promotion and application in the potential in high-latitude cold regions.

4. Conclusions

Air-NBW enhances methane production from anaerobic digestion of cattle manure under both mesophilic and psychrotrophic conditions by improving hydrolysis, optimizing microbial communities, and upregulating key functional genes involved in hydrogenotrophic and acetoclastic methanogenesis pathways. Air-NBW also reduces the total abundance of mobile genetic elements, particularly under mesophilic conditions. Furthermore, Air-NBW is associated with changes in the expression of functional genes involved in biofilm formation and antioxidant defense, which influence reactive oxygen species levels and, in turn, affect the potential for antibiotic resistance genes transfer via horizontal gene transfer. Under both mesophilic and psychrotrophic conditions, Air-NBW reduced the abundance of *lnu(D)*, *lsa(B)*, and *mph (B)* (MLS) as well as, *tet(W)*, *tet(T)*, and *tet(Q)* (tetracycline), thereby decreasing the total abundance of antibiotic resistance genes. Ultimately, Air-NBW reduced the ecological risk scores of the final digestate, with a particularly pronounced effect under psychrotrophic conditions.

Declaration of Generative AI and AI-assisted technologies in the writing process

The authors used ChatGPT to enhance the readability and language of the manuscript, and after review and edits, take full responsibility for the content of the published article.

All metagenomic sequencing raw files were uploaded to NCBI (accession numbers: PRJNA1447972, PRJNA1448473 and PRJNA1447957).

CRediT authorship contribution statement

Jiabei Shan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tianfeng Wang:** Writing – review & editing, Validation, Supervision, Resources, Funding acquisition. **Cheng Peng:** Validation, Investigation, Formal analysis. **Peilin Guo:** Investigation, Formal analysis. **Hong Zhu:** Investigation, Formal analysis. **Dan Luo:** Writing – review & editing, Methodology. **Yonggang Wang:** Resources. **Jixiang Chen:** Writing – review & editing, Validation, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China [grant numbers 52360019, 42267017].

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.134704.

Data availability

Data will be made available on request.

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